

Before starting, make sure you **read all the questions** and understand what is asked of you. Make sure you **justify your answers**.

1. (20 points) Translate the following C snippet into LLVM intermediate representation and into MIPS.

```
int trsum(int* array, int acc, int size) {
    int val = *array;
    if (size == 1)
        return acc + val;
    return trsum(array + 1, acc + val, size - 1);
}
```

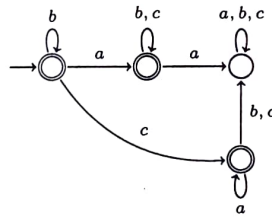
2. (4 points) Suppose we are given a *deterministic finite automaton* for a language $\mathcal{L}$ on Σ . For all words $w \in \Sigma^*$, we can use the automaton to determine whether $w \in \mathcal{L}$ in time:

- A. $\mathcal{O}(2^{|w|})$
- B. $\mathcal{O}(|w|^2)$
- C. $\mathcal{O}(|w|)$
- D. $\mathcal{O}(\log |w|)$

3. (6 points) According to the “longest match” principle, what are the first prefixes of

- (a) *abbbbc*
- (b) *abcba*
- (c) *bbbaa*
- (d) *bcabcaa*
- (e) *bbbcaab*

for which a scanner simulating the following automaton would produce a token?



Reminder

Recall that a grammar is LL(k) if and only if for all pairs of derivations

$$S \Rightarrow^* wA\gamma \Rightarrow^* w\alpha_1\gamma \Rightarrow^* wx_1$$

$$S \Rightarrow^* wA\gamma \Rightarrow^* w\alpha_2\gamma \Rightarrow^* wx_2$$

with $w, x_1, x_2 \in T^*$, $A \in V$, and $\gamma \in (V \cup T)^*$, if $\text{First}^k(x_1) = \text{First}^k(x_2)$ then $\alpha_1 = \alpha_2$.

A grammar is strongly LL(k) if and only if for all pairs of rules $A \rightarrow \alpha_1$ and $A \rightarrow \alpha_2$ with $\alpha_1 \neq \alpha_2$:

$$\text{First}^k(\alpha_1 \cdot \text{Follow}^k(A)) \cap \text{First}^k(\alpha_2 \cdot \text{Follow}^k(A)) = \emptyset.$$

4. (20 points) Consider the following grammar $\mathcal{G}$ and the language $\mathcal{L}$ it defines.

- (1) $S \rightarrow abc$
- (2) $S \rightarrow ac$
- (3) $S \rightarrow ba$
- (4) $S \rightarrow bc$

Which of the following are true? (Several may be true!)

- ☐ $\mathcal{G}$ is LL(1)
- ☐ $\mathcal{G}$ is strongly LL(2)
- ☐ $\mathcal{G}$ is LL(2)
- ☐ $\mathcal{L}$ is LL(3)

An interesting grammar

For the following two questions we will focus on the following grammar $\mathcal{G}$ and the language $\mathcal{L}$ it defines.

(1)	$S \rightarrow aA$	$\rightarrow a, aa, aaaa$
(2)	$S \rightarrow aB$	$\rightarrow aa$
(3)	$A \rightarrow \varepsilon$	
(4)	$A \rightarrow Aa$	
(5)	$B \rightarrow b$	
(6)	$B \rightarrow a$	

5. (20 points) Which of the following are true? (Several may be true!)

- ☐ $\mathcal{G}$ is LL(2)
- ☐ $\mathcal{G}$ is strongly LL(3)
- ☐ $\mathcal{L}$ is LR(1)
- ☐ $\mathcal{L}$ is LALR(1)

6. (20 points) The closure of $\{S \rightarrow \bullet aA, S \rightarrow a \bullet B\}$ includes:

- ☐ $S \rightarrow \bullet aA$
- ☐ $S \rightarrow a \bullet B$
- ☐ $A \rightarrow \bullet Aa$
- ☐ $B \rightarrow \bullet b$
- ☐ $B \rightarrow \bullet a$

7. (10 points) Let $(q, w, \gamma Z_0)$ be a configuration reached along a run ρ of the bottom-up parser with q not of the form $(A, \dots)$. The latter means that the parser is not busy popping elements as part of a reduce action. What can we conclude about ρ ?

- A. If $S \Rightarrow^* \gamma^R w$, that is the *reverse* of γ is a valid prefix of a sentential form, then ρ is an accepting run.
- B. If γ is not a valid prefix of a sentential form then ρ is not an accepting run.